DIBAKAR BARUA

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PROFESSIONAL SUMMARY

Machine Learning Engineer and Data Scientist with 3+ years of experience building and deploying ML/AI systems across NLP, Computer Vision, and predictive analytics. Proficient in Python, PyTorch, TensorFlow, and Scikit-learn, with end-to-end experience spanning data ingestion, feature engineering, model training, evaluation, and production deployment. Graduate researcher at the Visual Computing and Biometric Security Lab (UNT) applying deep learning to Autism Spectrum Disorder detection. Strong academic record (GPA 3.91, M.Sc. Data Science, University of North Texas). Targeting roles in Machine Learning Engineering, Data Science, Applied AI, or Data Analytics.

TECHNICAL SKILLS

Programming Languages: Python, SQL, C#, C++, Java, MATLAB

ML / Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Random Forest, LSTM, CNN, EfficientNet, ResNet, Transfer Learning

NLP & Generative AI: BERT, GPT, HuggingFace Transformers, RAG (Retrieval-Augmented Generation), LLM Fine-tuning, NLTK, SpaCy, Prompt Engineering

Computer Vision: OpenCV, EfficientNet-B0, ResNet-50, CNNs, Grad-CAM, Image Classification, Object Detection, Image Denoising

Data Science & Analytics: Pandas, NumPy, SciPy, Feature Engineering, Statistical Modeling, A/B Testing, Hypothesis Testing, Signal Detection

MLOps & Tools: Docker, Git, GitHub, Streamlit, REST APIs, Model Deployment, Jupyter Notebook, Google Colab, VS Code

Databases & Cloud: AWS (S3, EC2), MS SQL Server, MySQL, Oracle Database

Visualization: Tableau, Power BI, Matplotlib, Seaborn

PROFESSIONAL EXPERIENCE

Graduate Research Assistant — Computer Vision / Deep Learning *Aug 2025 – Present*

Visual Computing & Biometric Security Lab, University of North Texas | Denton, TX

- Conducting computer vision research on Autism Spectrum Disorder (ASD) detection using CNNs trained on behavioral and image-based datasets.
- Designing and evaluating deep learning architectures for ASD trait classification. Applying transfer learning, data augmentation, and Grad-CAM for model interpretability.
- Contributing to an active lab publication pipeline in collaboration with faculty and graduate researchers.

Student Assistant *Jun 2025 – Present*

University of North Texas | Denton, TX

- Developed skills in team coordination, event management, and time management in a high-traffic service environment.

Junior Machine Learning Engineering *Mar 2022 – Dec 2024*

Brain Station 23 | Dhaka, Bangladesh

- Designed and delivered end-to-end ML pipelines across 5+ client engagements spanning NLP, classification, and regression tasks using Python, PyTorch, and Scikit-learn.
- Cleaned and preprocessed datasets with 100K+ records, resolving class imbalance, missing values, and feature drift; improved baseline model accuracy by up to 15% through hyperparameter tuning and ensemble methods.
- Trained supervised learning models including Random Forest, XGBoost, LSTM, and BERT; presented results via Tableau and Matplotlib dashboards to non-technical stakeholders.
- Collaborated with software engineers, product managers, and domain experts to align model outputs with deployment requirements; documented architecture, metrics, and handoff procedures.

GIS & Data Visualization Intern *Sep 2021 – Jan 2022*

Bakhrabad Gas Distribution Co. Ltd. | Cumilla, Bangladesh

- Digitized 161 analog infrastructure maps and visualized spatial data for CNG stations and industrial facilities using QGIS under a government-sponsored digital transformation initiative.

Teaching Assistant — Object-Oriented Programming (C#) *Jun 2021 – Aug 2021*

American International University Bangladesh | Dhaka, Bangladesh

- Supported 80+ students on programming assignments in C#; held lab sessions covering Visual Studio, XAMPP, and GitHub.

MACHINE LEARNING & DATA SCIENCE PROJECTS

PhysicsDenoiseNet | *Deep Learning — Image Denoising* Jan - April *2026*

- Built a physics-informed deep learning model for real-world image denoising using PyTorch, with custom loss functions, dataset loaders, and server-ready training scripts.
- Trained and evaluated on SIDD Small, BSD68, and LOL benchmark datasets; assessed performance using PSNR, SSIM, LPIPS, and FID metrics.
- Generated CSV-based training logs, visualizations, and research-style LaTeX reports; analyzed failure modes including near-black output prediction on low-light inputs.

Pharmacovigilance Signal Detection — GLP-1 Receptor Agonists | *Health Informatics / Data Analytics* Jan - April *2026*

- Analyzed FDA FAERS adverse event reports (2015–2025) for GLP-1 receptor agonists; built a scalable data pipeline for large-scale drug safety analysis.
- Applied statistical signal detection methods (PRR, ROR) to identify potential psychiatric, gastrointestinal, and multi-organ adverse drug reaction signals.
- Cross-validated findings against FDA drug labeling and peer-reviewed safety references; demonstrated skills in health informatics, data cleaning, and evidence-based validation.

Clickbait Headline Detection | *NLP / Web App — University of North Texas* *Jan – Apr 2025*

- Built and compared Logistic Regression (TF-IDF), LSTM, and fine-tuned BERT classifiers for real-time clickbait detection; BERT achieved 99% classification accuracy.
- Deployed inference app via Streamlit with a REST API backend; managed full pipeline from preprocessing through evaluation (F1, precision, recall, AUC).

AI-Based Diabetic Retinopathy Detection | *Computer Vision — University of North Texas* *Jan – Apr 2025*

- Benchmarked EfficientNet-B0 and ResNet-50 for multi-class retinopathy grading from fundus images; EfficientNet-B0 reached 80% accuracy across 5 severity levels.
- Applied transfer learning, data augmentation, and Grad-CAM saliency maps to address class imbalance and improve clinical interpretability.

PUBLICATIONS & RESEARCH

Speaking at the Right Level: Literacy-Controlled Counter speech Generation with RAG-RL

EMNLP 2025| Published Aug 2025

- Developed an LLM-based counter speech generation system combining RAG and Reinforcement Learning to generate literacy-level-tailored responses to online hate speech; evaluated using Flesch-Kincaid readability, fluency, and relevance metrics. Published at EMNLP, a top-tier ACL venue.

Autism Spectrum Disorder (ASD) Prediction Using Machine Learning

IJITCS | Published Aug 2022

- Developed a hybrid SVM + CNN model achieving 94% accuracy on ASD trait prediction from behavioral datasets; integrated real-time Tableau dashboard via REST API for cross-cohort analysis.

EDUCATION

M.Sc. in Data Science *Expected Dec 2026*

University of North Texas | Denton, TX

GPA: 3.91 / 4.00 | Coursework: Machine Learning, Deep Learning, Health Informatics, Statistical Modeling, Big Data Analytics, Data Visualization

B.Sc. in Computer Science and Software Engineering *Graduated Aug 2021*

American International University Bangladesh | Dhaka, Bangladesh

GPA: 3.95 / 4.00 | Dean's Honors List, Summa Cum Laude